# Instructions for modelling primer

## Introduction

To support the Modelling Primer session we have provided a set of simple modelling exercises that introduce some key concepts common to much materials modelling practice. We will illustrate these concepts using classical molecular dynamics simulations. All of these simulations will be run via Google's Colab platform

If you haven't already got a google account, please sign up for one now.

## Setting up Google colab

The first step is to navigate to the github repository for the session

You should see something like in Figure 1.

Make sure you are viewing `ReadMe.md` and right-click the badge circled in the figure and select `open in new tab`. This will open a Colab tab and you should then see something like Figure 2.

Select `XX-CombinedExercises.ipynb`. You'll be asked a couple of questions. Confirm you want to proceed you'll then see the notebook in Colab as in Figure 3

If you click the file icon as shown in Figure 3 this will open your folder structure and you should see something like Figure 4.

The next step is to run the two code cells in this notebook. Click on the first code cell and then hit `shift + return`. You'll be asked to sign in to your Google account as in Figure 5.

Of the various permissions, you should only need to select those permissions that are obviously relevant to accessing your files on Google drive. You may need to hit the refresh button in the sidebar (Figure 6) but you should then see that `drive` now appears.

Execute the second code cell in the notebook (by clicking on it then hitting `shift + return`). This will clone the repository of files for the exercises into your Google drive. You should now

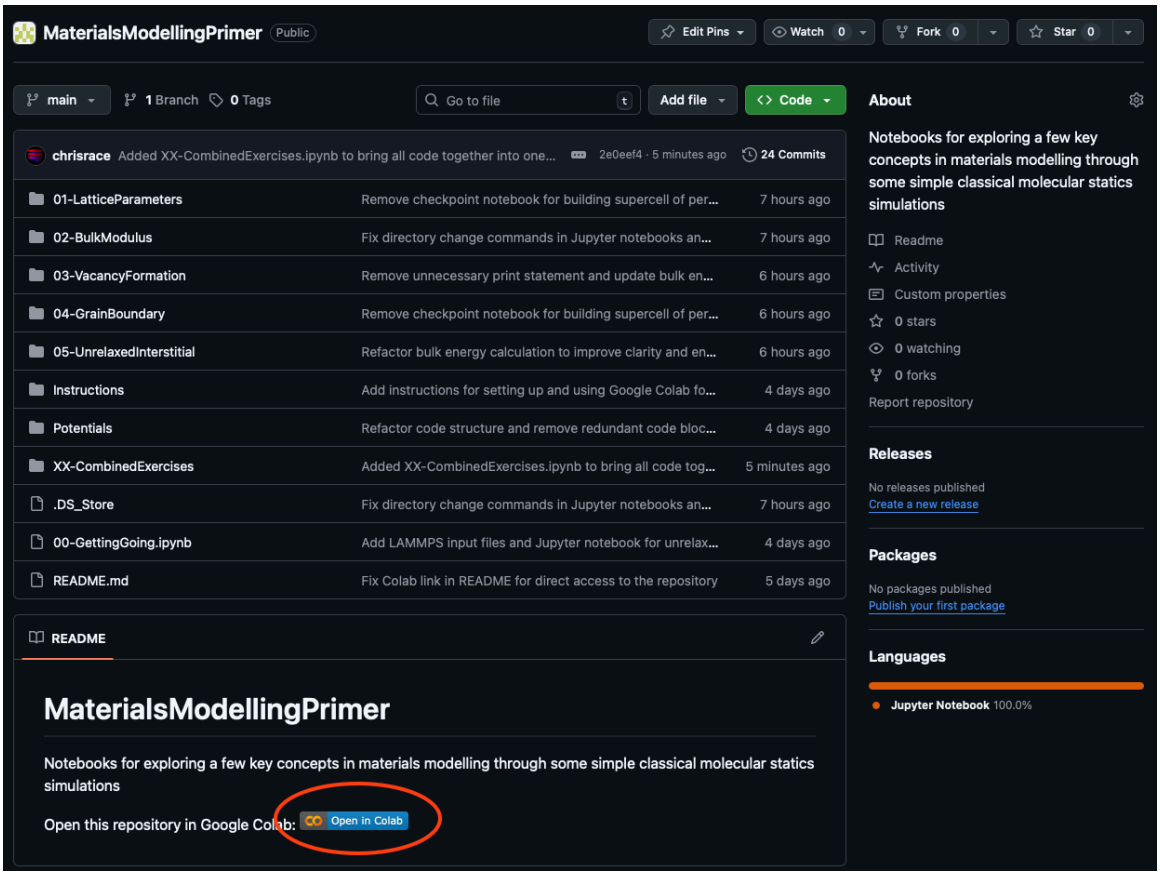
MaterialsModellingPrimer
Public
Edit Pins
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
t
Add file
Code
chrisrace Added XX-CombinedExercises.ipynb to bring all code together into one...
2e0eef4 · 5 minutes ago
24 Commits
01-LatticeParameters
Remove checkpoint notebook for building supercell of per...
7 hours ago
02-BulkModulus
Fix directory change commands in Jupyter notebooks an...
7 hours ago
03-VacancyFormation
Remove unnecessary print statement and update bulk en...
6 hours ago
04-GrainBoundary
Remove checkpoint notebook for building supercell of per...
6 hours ago
05-UnrelaxedInterstitial
Refactor bulk energy calculation to improve clarity and en...
6 hours ago
Instructions
Add instructions for setting up and using Google Colab fo...
4 days ago
Potentials
Refactor code structure and remove redundant code bloc...
4 days ago
XX-CombinedExercises
Added XX-CombinedExercises.ipynb to bring all code tog...
5 minutes ago
.DS_Store
Fix directory change commands in Jupyter notebooks an...
7 hours ago
00-GettingGoing.ipynb
Add LAMMPS input files and Jupyter notebook for unrelax...
4 days ago
README.md
Fix Colab link in README for direct access to the repository
5 days ago
README
MaterialsModellingPrimer
Notebooks for exploring a few key concepts in materials modelling through some simple classical molecular statics simulations
Open this repository in Google Colab:
Open in Colab
About
Notebooks for exploring a few key concepts in materials modelling through some simple classical molecular statics simulations
Readme
Activity
Custom properties
0 stars
0 watching
0 forks
Report repository
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Languages
Jupyter Notebook 100.0%

Figure 1

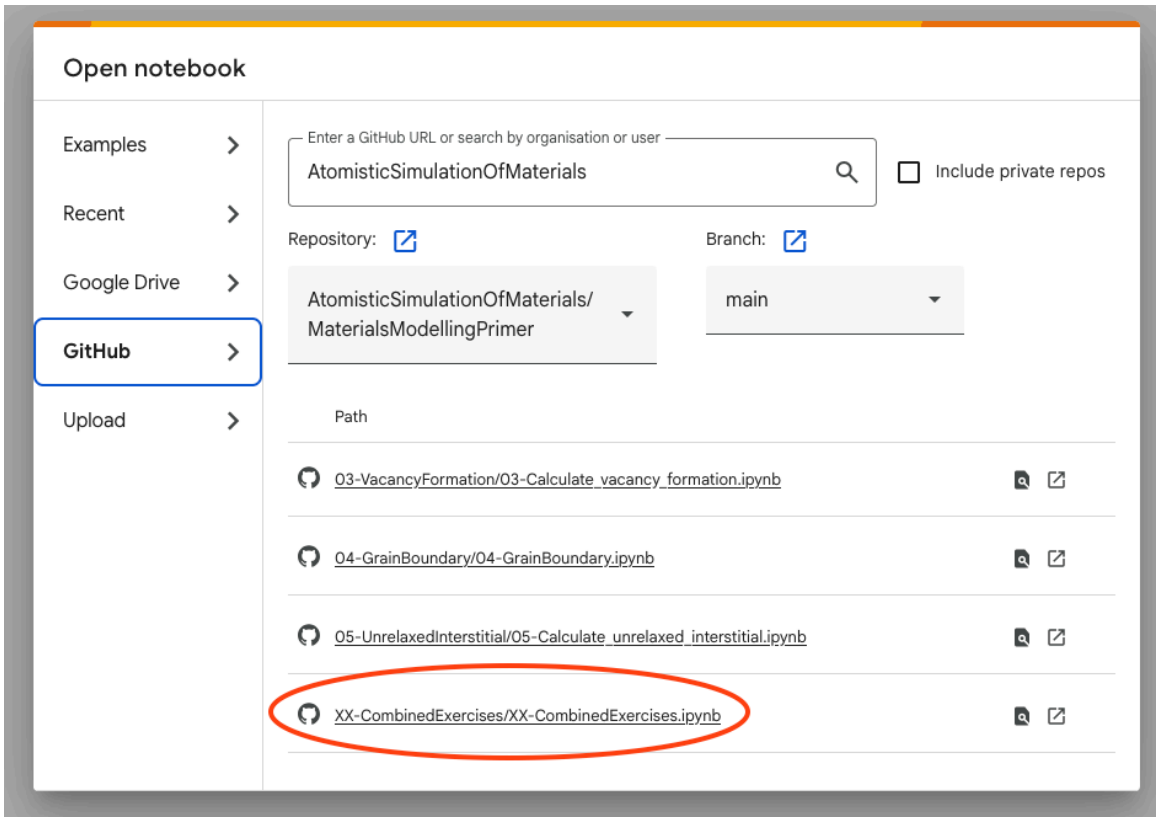
Open notebook
Examples
Recent
Google Drive
GitHub
Upload
Enter a GitHub URL or search by organisation or user
AtomisticSimulationOfMaterials
Include private repos
Repository:
Branch:
AtomisticSimulationOfMaterials/
MaterialsModellingPrimer
main
Path
03-VacancyFormation/03-Calculate_vacancy_formation.ipynb
04-GrainBoundary/04-GrainBoundary.ipynb
05-UnrelaxedInterstitial/05-Calculate_unrelaxed_interstitial.ipynb
XX-CombinedExercises/XX-CombinedExercises.ipynb

Figure 2

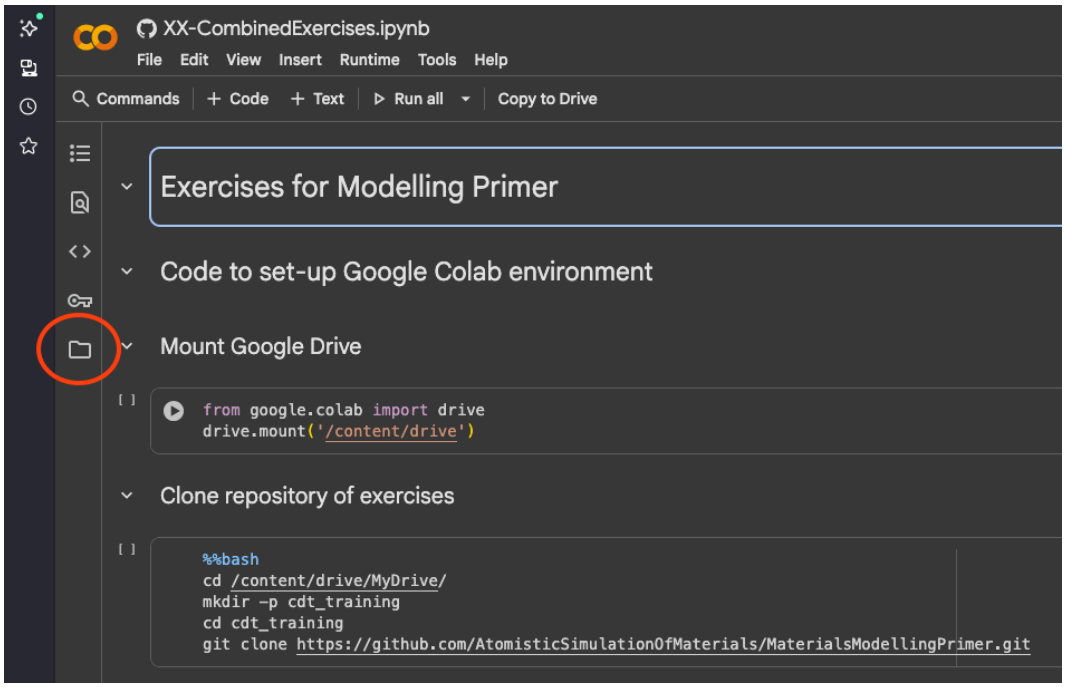


Figure 3

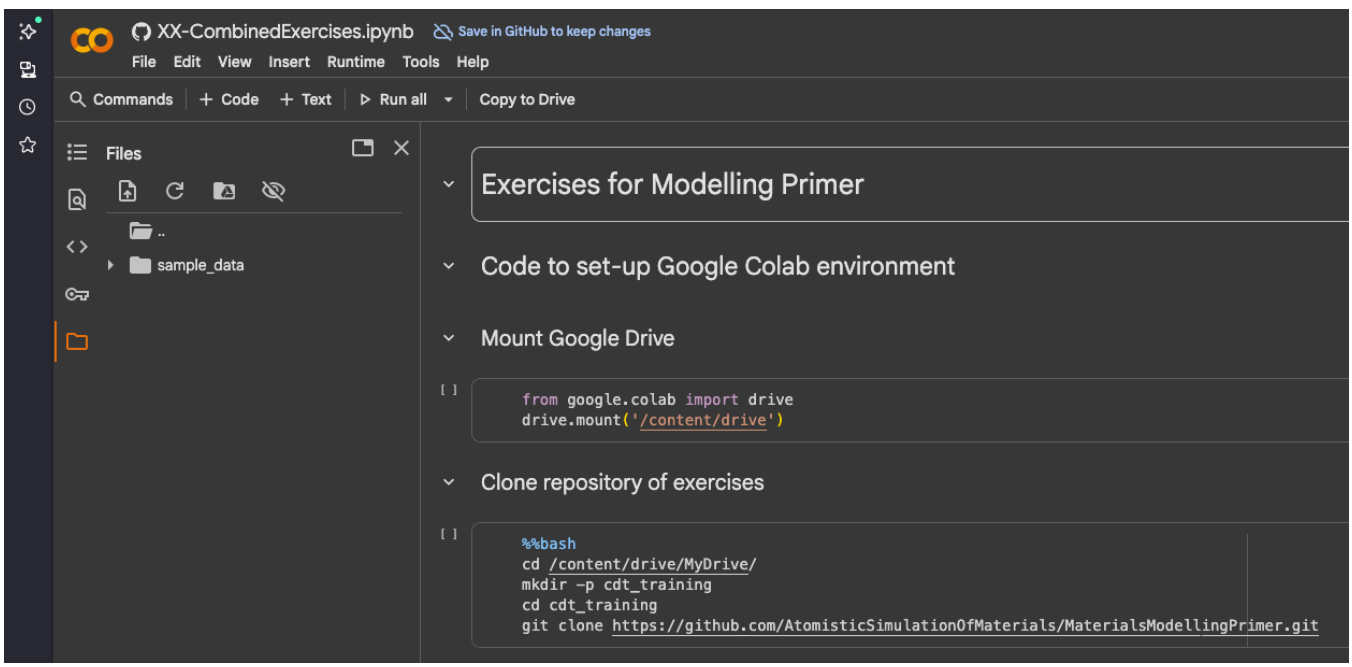


Figure 4

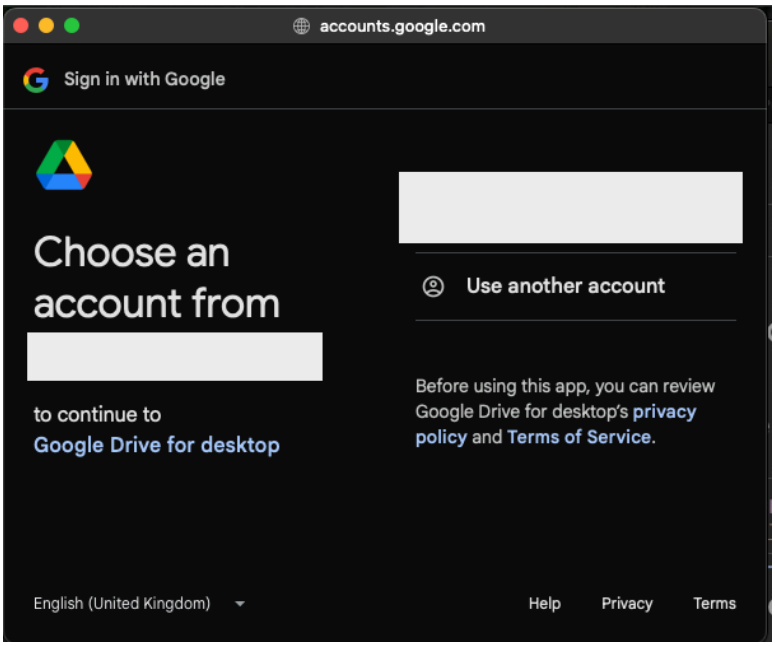


Figure 5

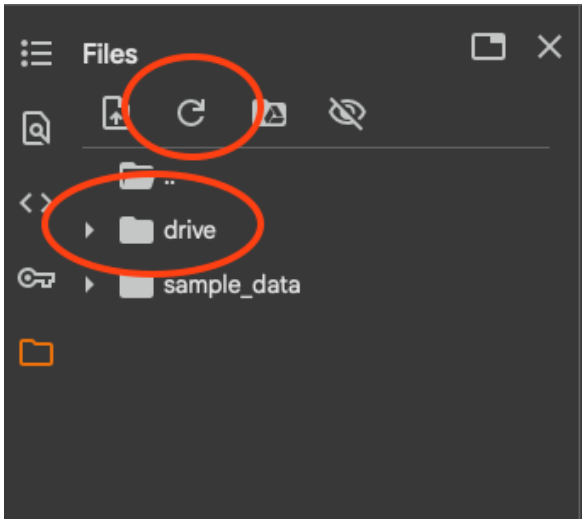


Figure 6

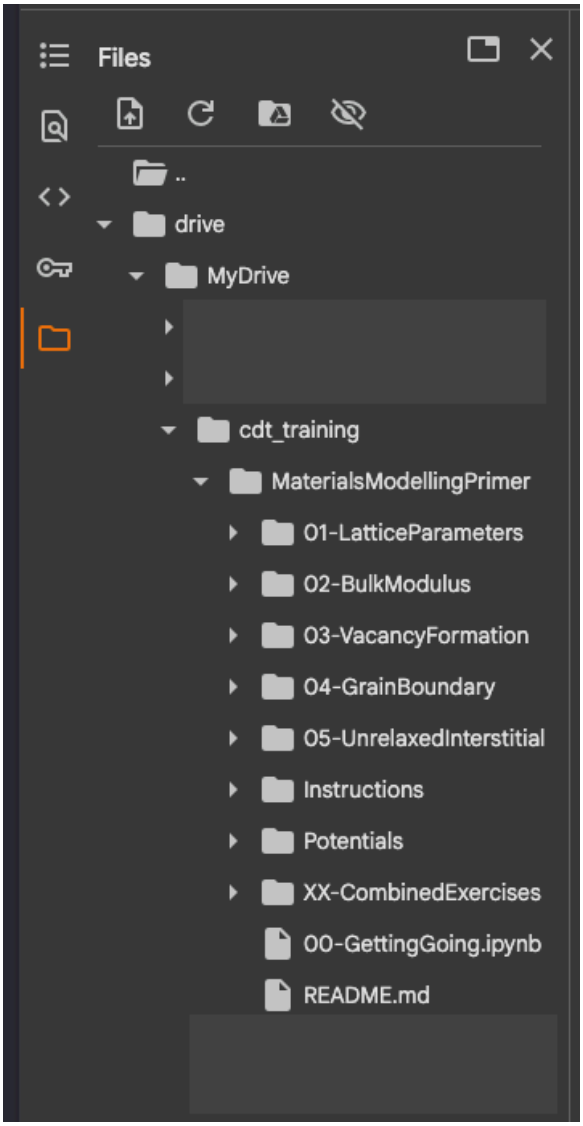
Files
..
drive
MyDrive
cdt_training
MaterialsModellingPrimer
01-LatticeParameters
02-BulkModulus
03-VacancyFormation
04-GrainBoundary
05-UnrelaxedInterstitial
Instructions
Potentials
XX-CombinedExercises
00-GettingGoing.ipynb
README.md

Figure 7

be able to find all the files for the session under `drive/MyDrive` as in Figure 7 (my Google drive also has a lot of clutter in it).

Run the next three code cells (see Figure 8). These cells:

- Install the lammps molecular dynamics software (and a module that will help us read the output)
- Load the python modules that we will need for the exercises
- Define a function we will us a lot in the exercises.

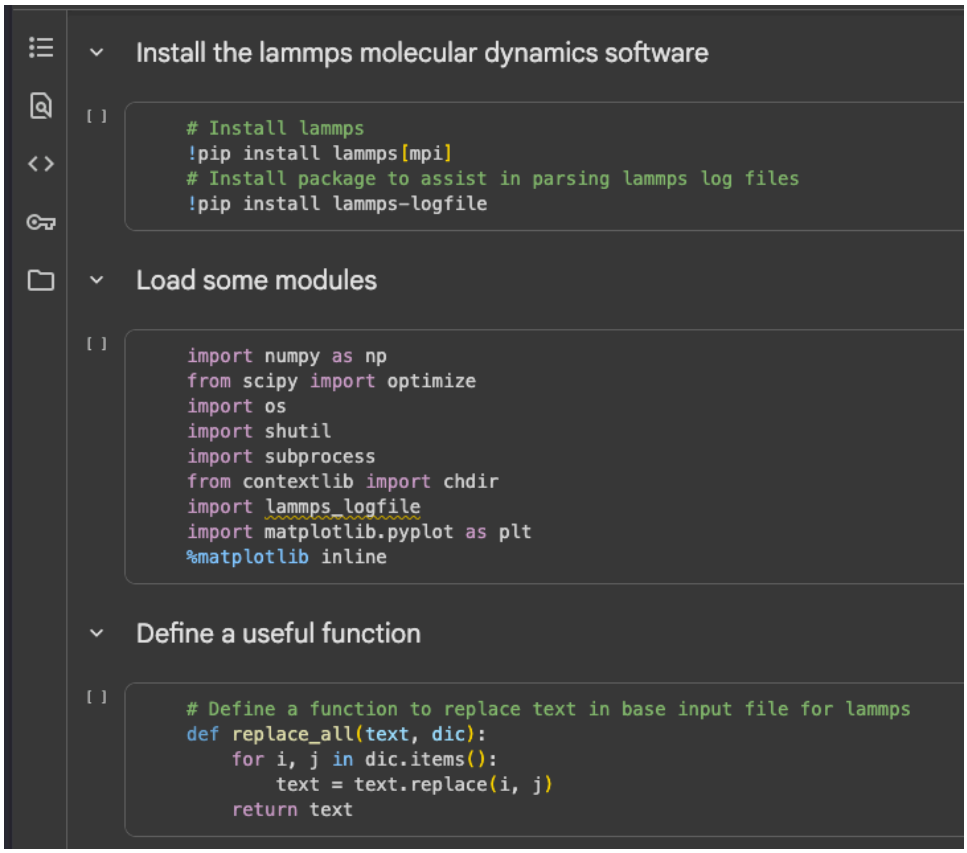


Figure 8

## The exercises

All that ermains now is to work through the exercises in the rest of the notebook. I'll let you know the right point in the lecture to run each exercise.

Note that the exercises will be run in the folders:

- `02-BulkModulus/`
- `03-VacancyFormation/`
- `04-GrainBoundary/`
- `04-GrainBoundary/`
- `05-UnrelaxedInterstitial/`

You can ignore the notebooks in these folders (they reproduce the contents of the current notebook, `XX-CombinedExercises.ipynb`, but for one exercise at a time).